

03_FOG_GOVERNOR_RESULTS

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EVIDENCE PROVENANCE: [INJECTED + HARDWARE LOOP]

Evaluated via Weather Station API fog injection through authoritative Twin to hardware motor actuation.

1. Monotonic Fog Response Matrix

When weather station visibility drops, the governor scales safe speed monotonically. Central dispatch requests are clamped unconditionally when exceeding safe speed.

Fog Intensity	Condition	Visibility (m)	Fog Factor	v_safe (m/s)	Governor State
0.00	CLEAR	1000 m	1.00	0.80 m/s	NORMAL (At road ceiling)
0.25	LIGHT FOG	500 m	0.75	0.60 m/s	ACTIVE - FOG CONSTRAINT
0.50	MODERATE FOG	250 m	0.50	0.40 m/s	ACTIVE - FOG CONSTRAINT
0.75	HEAVY FOG	100 m	0.30	0.24 m/s	ACTIVE - FOG CONSTRAINT
1.00	SEVERE FOG	30 m	0.10 / 0.00	0.08 / 0.00 m/s	SEVERE STOP (Crawl / Hold)
Recovery	CLEAR	1000 m	1.00	0.80 m/s	NORMAL (No hysteresis)

2. Monotonic Non-Increasing Invariant

The derivative dv_safe / dF_fog is strictly positive; increasing fog unconditionally lowers safe speed. Recovery occurs smoothly with zero velocity spikes.